



THE UNIVERSITY
of EDINBURGH



Biotechnology and
Biological Sciences
Research Council



THE ROYAL
SOCIETY

Maize Program

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Minneapolis, June 2026



Bayer CropScience

Overview

Performs a simulation of a maize breeding program to measure the effect of switching from initial evaluation of DH lines on testcross performance to evaluation on per se performance. The simulation models 20 years of burn-in for a breeding using the testcross scheme followed by 20 years of breeding in either the testcross or per se scenarios. The design of the breeding program is based on the breeding program on page 211 of "Essentials of Plant Breeding". The rationale for this comparison is based on a recommendation of Troyer and Wellin (2009). The number of QTL and the values for dominance degree have been chosen to approximately match the long-term rates of genetic gain shown in figure 2 of Troyer and Wellin (2009).

Bernardo, R. 2014. Essentials of Plant Breeding. Stemma Press, Woodsbury, MN.

Troyer, A.F., and E.J. Wellin. 2009. Heterosis Decreasing in Hybrids: Yield Test Inbreds. Crop Science 49(6): 1969–1976.

Breeding Program

- Year 1
 - Making 80 crosses from 50 parents per heterotic pool
 - Parents come from 2nd through 4th years of yield trial testing
 - Making 50 DH lines per cross
- Year 2
 - Crossing each DH line to one tester and evaluating hybrid at one location
 - Selecting 400 best DH lines per heterotic pool
- Year 3
 - Crossing each DH line to three testers and evaluating at two locations
 - Selecting 40 best DH lines per heterotic pool based on GCA
- Year 4
 - Crossing each DH line to 5 elite lines from the opposing heterotic pool and evaluating at 4 locations
 - Selecting the 20 best hybrids per heterotic pool
- Year 5
 - Reevaluating selected hybrids at 8 locations and selecting the 4 best per heterotic pool
- Year 6
 - Reevaluating selected hybrids in strip trials before release

Running the Simulation

- Step 1: run BURNIN.R
 - Models 20 years in the standard (testcross) breeding program
- Step 2: run BASELINE.R
 - Models 20 additional years in the standard breeding program
- Step 3: run PERSE.R
 - Models 20 additional years (after burn-in) in a breeding program that substitutes the first test cross yield trial for *per se* testing of inbreds
- Step 4: run ANALIZERESULTS.R
 - Generates plots of the output